ShieldNet AI: Customer Solution Brief

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1. Executive Summary

ShieldNet AI is a real-time malicious URL detection platform that combines deterministic analysis, semantic vector retrieval, Atlas text search, and optional Foundry-based agentic reasoning.

It helps security teams triage suspicious URLs faster with explainable evidence and campaign-level context.

2. Business Value

- Accelerates URL triage through automated risk scoring and evidence cards.
- Improves detection quality using lexical + semantic retrieval.
- Supports campaign intelligence by clustering related malicious URLs.
- Enables analyst workflows using threat database filters and status operations.

3. High-Level Architecture

Frontend: Next.js 15 + React 18 + Recharts

Backend: FastAPI + Motor + Pydantic

Data & Search: MongoDB Atlas (Atlas Search + Vector Search + Hybrid Rank Fusion)

Embeddings: Voyage API (voyage-4 default, 1024 dimensions)

Optional Agentic Layer: Microsoft Foundry (gpt-5.4 default)

4. Key Functional Modules

- URL Scanner: extraction, scoring, vector evidence, recommendation.

- Threat Database: searchable repository with operational triage.

- Search Playground: Atlas vs Vector vs Hybrid comparison.

- Campaign Detection: groups related threats and tags URLs.

- Rules Management: active threshold and weight configuration.

- URL Graph: relation edges between similar threats.

5. Data Sources and Ingestion

- Live scan ingestion writes/updates records in urls collection.

- Threat-intel seed writes threat docs into urls using docType=threat_intel.

- Campaign detection writes campaign aggregates in campaigns.

- Similarity graph writes edges into url_edges.

- Active policy config is maintained in scan_rules.

6. Sample Schemas

6.1 urls (scan record)

```
{
  "url": "https://secure-login-incometax.xyz/auth/verify",
```

```
  "domain": "secure-login-incometax.xyz",
  "canonicalDomain": "incometax.gov.in",
  "docType": "scan",
  "threatClassification": "phishing",
  "riskScore": 86.4,
  "status": "blocked",
  "summaryText": "Credential harvest style URL impersonating tax portal...",
  "embedding": ["1024-dim vector"],
  "payloadTypes": ["credential_harvest", "open_redirect"],
  "campaignId": "campaign_001_tax_auth_2025",
  "scanCount": 3,
  "firstSeenAt": "2026-04-27T08:51:00Z",
  "lastSeenAt": "2026-04-30T11:08:12Z"
}
```

6.2 urls (threat-intel record in same collection)

```
{
  "url": "https://nic.in/login?redirect=http://gov-login-verify.xyz/capture",
  "domain": "nic.in",
  "docType": "threat_intel",
  "threatClassification": "phishing",
  "attackCategory": "open_redirect",
  "feedName": "CERT-IN_Feed",
  "summaryText": "Open redirect attack against nic.in ...",
  "embedding": ["1024-dim vector"]
}
```

6.3 campaigns

```
{
  "campaignId": "campaign_001_tax_auth_2025",
  "name": "Tax Authority Phishing Campaign Q1 2025",
  "attackCategory": "credential_harvest",
  "status": "active",
  "urlCount": 128,
  "avgRiskScore": 83.7,
  "domains": ["incometax.gov.in", "gst.gov.in"],
  "lastSeen": "2026-04-30T10:21:00Z"
}
```

6.4 url_edges

```
{
  "fromUrl": "https://secure-login-incometax.xyz/auth/verify",
  "toUrl": "https://gst-auth-update.top/kyc/verify",
  "strength": 0.79,
  "factors": [
    {"name": "vector_similarity", "value": 0.84},
    {"name": "shared_attack_category", "value": 0.74}
  ]
}
```

6.5 scan_rules

```
{
  "_id": "active_rules",
  "thresholds": {"block": 70, "review": 45},
  "weights": {"vector": 0.30, "payload": 0.20, "dga": 0.20}
}
```

7. Embedding Strategy

- Embed summaryText (URL + metadata + threat signals), not raw URL alone.

- Default model: voyage-4; vector size: 1024; similarity: cosine.

- Campaign-aware re-embedding improves future threat/campaign retrieval precision.

8. Detection Flow

1) Input URL -> 2) Feature extraction -> 3) Summary + embedding -> 4) Vector search -> 5) Risk engine -> 6) Campaign tag/update -> 7) Persist + response

9. Operational Notes

- This is a validation/demo deployment and should be calibrated for production traffic.

- Keep secrets in environment variables and enforce role-based access on rules/status overrides.

Prepared for customer sharing. Customize with project branding, SLAs, and rollout plan.